MIL-S-10145D(ME)
15 August 1978
SUPERSEDING
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22 April 1965

MILITARY SPECIFICATION

SKETCHMASTER, VERTICAL, WITH CASE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of vertical sketchmaster, complete with case.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel, and Seals.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-621	- Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation-Packaging, Methods of.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.

DRAWINGS

ME

D5933-1 through -15	- Sketchmaster, Vertical, with Case.

(Copies of specifications, standards, and drawings required by contractors in connection with specific procurement functions shall be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. The vertical sketchmaster with case shall be as shown on Drawings D5933-1 to -15 inclusive and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the sketchmaster. Any data (e.g. shop drawings, layouts, flow sheets,

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processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (first produced sketchmaster with case). The contractor shall furnish one sketchmaster with case as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods will produce sketchmasters with cases that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.5).

3.4 Mirrors. The mirrors shall be of such optical flatness that no pair of interference fringes within the area of the optical flat .250 inch (6.35 mm) from the edges shall be separated by less than .079 inch (2 mm). The reflecting surfaces of the mirrors shall be clean and free from cracks and shall show no scratches or pinholes when subjected to salt solutions or when rubbed with a soft cloth.

3.4.1 Semitransparent mirror. A combination of film material having alternatingly high- and low-refraction indices shall be used. The separate films of the coatings shall be applied by vacuum evaporation to such a thickness that the mirror shall provide a reflection of light in the visible spectrum of 70 percent, plus or minus 4 percent, and a transmission of light in the visible spectrum of 30 percent, plus or minus 4 percent. The absorption of visible light (the difference between reflection and transmission values) shall be not more than 4 percent. The opposite faces of the mirror shall be parallel to within 5 minutes of arc.

3.4.2 Fully reflecting mirror. The fully reflecting mirror shall have all edges ground, and the front shall be surfaced with aluminum by vacuum evaporation so that reflection of light in the visible spectrum shall be not less than 85 percent. The reflective layer shall be protected by a coating of silicon monoxide deposited to a thickness of approximately 1,500 Angstrom units.

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3.5 Lenses. The lenses shall be ophthalmic lenses ground from optical glass and highly polished. Each lens shall show a ratio range in relationship to its lens height above the table and its diopter power as indicated in Table I.

TABLE I. Powers and focal lengths of lenses and height settings for the vertical sketchmaster.

Height - lens to table		Lens power in diopters (D) and focal length (f)			Ratio range
(inches)	(mm)	D	f (inches)	(mm)	
9.5 to 13.5	241. to 343.	-	-	-	0.86 to 1.14
7.6 to 10.1	193. to 257.	1	40	1016.0	0.69 to 0.86
6.2 to 8.0	157. to 203.	2	20	508.0	0.57 to 0.69
5.3 to 6.8	135. to 173.	3	13	330.2	0.49 to 0.57
4.6 to 5.8	117. to 147.	4	10	254.0	0.42 to 0.49
4.1 to 5.0	104. to 127.	5	8	203.2	0.38 to 0.42
3.6 to 4.5	91. to 114.	6	6	152.4	0.33 to 0.38
3.0 to 4.2	76. to 107.	8	5	127.0	0.28 to 0.33
2.8 to 3.5	71. to 89.	10	4	101.6	0.24 to 0.28

3.6 Identification marking. The sketchmaster shall be identified in accordance with MIL-STD-130.

3.7 Instruction plate. An instruction plate with the following information shall be fastened on the inside cover of the case in a location where the plate will be both visible and legible.

"NOTICE. THE USE OF COTTON AND WATER OR A LIQUID CLEANSER, SUCH AS AMMONIA WATER, IS RECOMMENDED FOR CLEANING THE MIRRORS OF THIS INSTRUMENT. USE EXTREME CAUTION TO PREVENT DAMAGE TO THE MIRRORS."

The instruction plate shall conform to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material. The plate shall be attached by screws, bolts, or rivets.

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3.8 Treatment and painting. The portions of the sketchmaster normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, Type A.

3.9 Workmanship. The quality of workmanship shall be such as to produce sketchmasters with cases that are in accordance with the requirements of this specification and are so constructed as to insure proper functioning of all parts of the units. All parts, components, and assemblies of the sketchmaster including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces, and welded parts shall be clean and free from sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All edges shall be rounded or beveled.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.1.2 Disassembly inspection. Failure of any examination or test by the first produced sketchmaster with case shall be cause for disassembly, in the presence of a Government representative, of the first produced sketchmaster with case to the extent necessary to determine the cause of the failure. Each disassembled part shall be examined in detail for compliance with this specification and referenced drawings in regard to materials, dimensions, tolerances, and workmanship. Parts not complying with such requirements shall be rejected and shall be cause for rejection of the first produced sketchmaster with case. Reassembly with replacement parts and retesting shall be the responsibility of the contractor.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First produced sketchmaster with case inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of packaging (see 4.6).

4.3 First produced sketchmaster with case inspection.

4.3.1 Examination. The first produced sketchmaster with case shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.3.2 Tests. The first produced sketchmaster with case shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection or for performing the disassembly inspection specified in 4.1.2.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2. AQL shall be 2.5 percent defective.

4.5 Inspection procedure.

4.5.1 Examination. The sketchmaster with case shall be examined as specified herein for the following defects:

Major

101. Dimensions not as specified.
102. Components missing or not as specified.
103. Material not as specified.
104. Instruction plate not as specified.
105. Treatment and painting not as specified.
106. Workmanship not as specified.

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Minor

201. Identification marking missing or illegible.

4.5.2 Tests.

4.5.2.1 Mirror flatness. The mirrors shall be placed on a flat surface and illuminated with a mercury light of 5,460 Angstrom units and adjusted until a minimum number of fringes is evident. Evidence of a pair of interference fringes within the area of the optical flat .250 inch (6.35 mm) from the edges separated by more than .079 inch (2 mm) shall constitute failure of this test.

4.5.2.2 Mirror corrosion-resistance. The mirrors shall be immersed in a 6 percent salt solution (NaCl) at room temperature for not less than 12 hours, removed from the solution, and washed. Evidence of cracks, scratches, pinholes, or any damage shall constitute failure of this test.

4.5.2.3 Mirror abrasion-resistance. A .750 inch (19.05 mm) steel ball, covered with linen having a thread count of not more than 60 threads per inch, shall be loaded with a 500 gram weight and rubbed on the reflecting surface of the mirrors with a reciprocating motion for not less than 100 cycles. Evidence of cracks, scratches, pinholes, or any damage shall constitute failure of this test.

4.5.2.4 Semitransparent mirror. The semitransparent mirror shall be tested for light transmission and light reflection in the visible spectrum. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5.2.5 Mirror reflection. The fully reflecting mirror shall be tested for reflection of light in the visible spectrum. Light reflection of less than 85 percent shall constitute failure of this test.

4.5.2.6 Lens focal length. The focal length and the magnification of each lens shall be checked. Inability of the lens to meet the requirement in Table I shall constitute failure of this test.

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4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials or method not as specified. Each incorrect material and each incorrect method shall constitute one defect.
108.	Box closure for Level A preservation not Method V as specified.
109.	Strapping not as specified.
110.	Boxes not reinforced for Level B.
111.	Cushioning not provided as specified for Level A and Level B.
112.	Marking illegible, incomplete, incorrect, or missing.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each sketchmaster and the technical publications contained in the case shall be preserved in accordance with MIL-P-116, Method IIb, with the required amount of desiccant placed in the case. The case shall be used as the inner container and a close-fitting box conforming to PPP-B-636, W5c or W6c, shall be used as the outer container. Box closure shall be in accordance with the appendix to the box specification, Method V.

5.1.2 Commercial. The sketchmasters shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be Level A, Level B, or Commercial, as specified (see 6.2).

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5.2.1 Level A. Sketchmasters with case, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-621, Class 2, style optional; or PPP-B-601, Overseas Type, style optional. Cushioning shall be provided to prevent shifting of contents. Strapping shall conform to QQ-S-781, Class 1, Type I or IV, size as applicable. Unless otherwise specified (see 6.2), strapping shall be Finish A.

5.2.2 Level B. Sketchmasters with case, preserved as specified in 5.1, shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, style optional, or PPP-B-640, Class 2, style optional. Cushioning shall be provided to prevent shifting of contents. The boxes shall be reinforced with bands of reinforced tape, applied in accordance with the appendix to the applicable box specification.

5.2.3 Commercial. Sketchmasters shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military packaging. Marking shall be in accordance with MIL-STD-129.

5.3.2 Commercial packaging. Marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The sketchmaster with case is intended for use in compilation of planimetric detail from 9- by 9-inch (228.6 by 228.6 mm), near-vertical, aerial photographs with a range of plotting scales from 0.25 to 1.25 times the scale of the photograph, on a base sheet or map manuscript by simultaneously viewing both the aerial photograph and the base sheet or map manuscript.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of a first produced sketchmaster (see 3.2).
(c) Degree of preservation and degree of packing required (see 5.1 and 5.2).
(d) When strapping shall be other than specified (see 5.2.1).

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6.3 First produced sketchmaster with case. Any changes or deviations of production sketchmasters with cases from the approved first produced sketchmaster with case during production will be subject to the approval of the contracting officer. Approval of the first produced sketchmaster with case will not relieve the contractor of his obligation to furnish sketchmasters with cases conforming to this specification.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each sketchmaster with case.

6.5 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

6.6 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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